Outline

1. Motivation
2. Basic Ideas and Terminology
3. Solutions of Differential Equations
4. Initial-Value Problem
5. Separable Differential Equations
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7. First-Order Linear Differential Equations
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Exact Differential Equations

The technique presented in this section is for first-order differential equations written in a differential form

$$M(x, y)\, dx + N(x, y)\, dy = 0.$$

Recall from a **calculus course** that if $u = u(x, y)$ is a function of two variables x and y, then its **differential** du is defined by

$$du = \frac{\partial u}{\partial x}\, dx + \frac{\partial u}{\partial y}\, dy.$$

Example

Solve the equation

$$2x \sin y \, dx + x^2 \cos y \, dy = 0. \tag{1}$$

Solution

This equation is separable, but we will use a different technique to solve it.

We notice that $2x \sin y \, dx + x^2 \cos y \, dy = d(x^2 \sin y)$. Then Equation (1) can be written as $d(x^2 \sin y) = 0$, which implies that $x^2 \sin y$ is constant, denoted by C. Thus the general solution to Equation (1) is

$$\sin y = \frac{C}{x^2},$$

where C is an arbitrary constant.

□

Obviously, in the example above, it was able to write the given differential equation in the form $du(x, y) = 0$, and hence obtain its solution.

However, we cannot always do it. Indeed, we see that the differential equation

$$M(x, y)\, dx + N(x, y)\, dy = 0$$

can be written as $du = 0$ **if and only if**

$$M = \frac{\partial u}{\partial x} \quad \text{and} \quad N = \frac{\partial u}{\partial y}$$

for some function u.

This leads to the following definition:

Definition

The differential equation

$$M(x, y)\, dx + N(x, y)\, dy = 0 \tag{2}$$

is said to be **exact** in a domain **D** of the xy-plane if there exists a function $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = M, \ \frac{\partial u}{\partial y} = N, \tag{3}$$

for all $(x, y) \in \mathbf{D}$.

Any function u satisfying (3) is called a **potential function** for the Equation (2).

If such a function does exist, then Equation (2) can be written as

$$\begin{aligned} & M(x,y)\,dx + N(x,y)\,dy = 0 \\ \iff & \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = 0 \\ \iff & du = 0. \end{aligned}$$

This is a reason why such a differential equation is called an **exact** differential equation.

For the example above, the potential function is

$$u(x,y) = x^2 \sin y.$$

The following result shows that if a differential equation is exact and if a potential function u can be found, then we can write down its solution immediately.

Theorem

The general solution to an exact differential equation

$$M(x, y)\, dx + N(x, y)\, dy = 0$$

is defined **implicitly** *by*

$$u(x, y) = C, \tag{4}$$

where $u(x, y)$ satisfies (3) and C is an arbitrary constant.

Proof

Exrecise.

There naturally arise 2 questions:

1. **How to verify if a given differential equation is exact?**
2. **How to find a potential function for an exact equation?**

The answer is given in the following theorem:

Theorem (Test for Exactness)

Let M, N, and their first partial derivatives M_y and N_x, be **continuous** *in a* ***simply connected*** *domain* ***D*** *of the xy-plane. Then the differential equation*

$$M(x,y)dx + N(x,y)dy = 0$$

is **exact** *for all (x,y) in* ***D*** **if and only if**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \text{ that is } M_y = N_x.$$

Here a **simply connected** domain **D** is a domain such that every simple closed contour within it encloses only points of **D**.

- The set of points interior to a simple closed contour is an example.
- The annular domain between two concentric circles is not simply connected.

A domain that is not simply connected is said to be *multiply connected*.

Remark

Since there is **no need** to memorize the final result for u, we omit the details of the proof of the theorem above.

For a particular problem one can construct an appropriate potential functions from first principles.

Examples

Determine whether the given differential equation is exact.

1. $[1 + \log(xy)]\, dx + \frac{x}{y}\, dy = 0.$
2. $x^2 y\, dx - (xy^2 + y^3)\, dy = 0.$

Solution

1. We have $\begin{cases} M(x,y) = [1 + \log(xy)] \implies M_y = \frac{1}{y} \\ N(x,y) = \frac{x}{y} \implies N_x = \frac{1}{y}. \end{cases}$
   So $M_y = N_x$ and the differential equation is exact.
2. We have $\begin{cases} M(x,y) = x^2 y \implies M_y = x^2 \\ N(x,y) = -(xy^2 + y^3) \implies N_x = -y^2. \end{cases}$
   Thus $M_y \neq N_x$, and hence the differential equation is **not** exact.

Example

Find the general solution to

$$2xe^y\ dx + (x^2e^y + \cos y)\ dy = 0.$$

Solution

We have

$$M(x,y) = 2xe^y,\ N(x,y) = x^2e^y + \cos y \implies M_y = 2xe^y = N_x.$$

Thus the given differential equation is **exact**, and so there exists a potential function u such that

$$\frac{\partial u}{\partial x} = 2xe^y, \tag{5}$$

$$\frac{\partial u}{\partial y} = x^2e^y + \cos y. \tag{6}$$

Integrating Equation (5) w.r.t. x, keeping y fixed, we have

$$u(x, y) = x^2 e^y + \mathbf{g(y)}, \tag{7}$$

where $g(y)$ is an arbitrary function of y.

We now determine $g(y)$ so that (7) also satisfies Equation (6). Taking the derivative of (7) w.r.t y yields

$$\frac{\partial u}{\partial y} = x^2 e^y + \frac{dg}{dy}. \tag{8}$$

So we obtain Equations (6) and (8) which give two expressions for $\frac{\partial u}{\partial y}$. This allows us to determine $g(y)$.

We have

$$\begin{cases} \dfrac{\partial u}{\partial y} = x^2 e^y + \cos y & (6) \\ \dfrac{\partial u}{\partial y} = x^2 e^y + \dfrac{dg}{dy}. & (8) \end{cases}$$

Subtracting Equation (6) from Equation (8), we get

$$\frac{dg}{dy} = \cos y,$$

which implies that

$$g(y) = \sin y,$$

where we have set the integration constant equal to zero w.l.o.g., since we require only **one** potential function.

Substitution $g(y) = \sin y$ into (7)

$$u(x, y) = x^2 e^y + g(y) \quad (7)$$

yields the potential function

$$u(x, y) = x^2 e^y + \sin y.$$

Consequently, the given differential equation can be written as

$$d(x^2 e^y + \sin y) = 0,$$

and hence, the general solution is

$$x^2 e^y + \sin y = C,$$

where C is an arbitrary constant.

□

Discussion.

In the solution above, we started our working with Equation (5) first, by integrating it w.r.t. x, keeping y fixed

$$\frac{\partial u}{\partial x} = 2xe^y.$$

A question to ask is: What happens, if we start with Equation (6) first

$$\frac{\partial u}{\partial y} = x^2e^y + \cos y,$$

and what the solution is then?

This is left as an exercise (!)

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Numerical Solution to First-order Differential Equations

So far in this chapter we have investigated first-order differential equations analytically by trying to construct exact solutions to certain types of differential equations.

Certainly, for most first-order differential equations, it simply is not possible to find analytic solutions, since they will not fall into a few classes for which solution techniques are available.

Our final approach to analyzing first-order differential equations is to look at the possibility of constructing a numerical approximation to the unique solution to the initial-value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \tag{1}$$

The idea is to generate a sequence of approximations $y_1, y_2, \ldots$ to the value of the exact solution at the points $x_1, x_2, \ldots$, where $x_{n+1} = x_n + h$ $n = 0, 1, 2, \ldots$, and h is a real number .

Remark.

- Numerical methods don't generate a formula for the solution to differential equation.
- Rather they generate a sequence of approximations to the values of the solution at specific points.

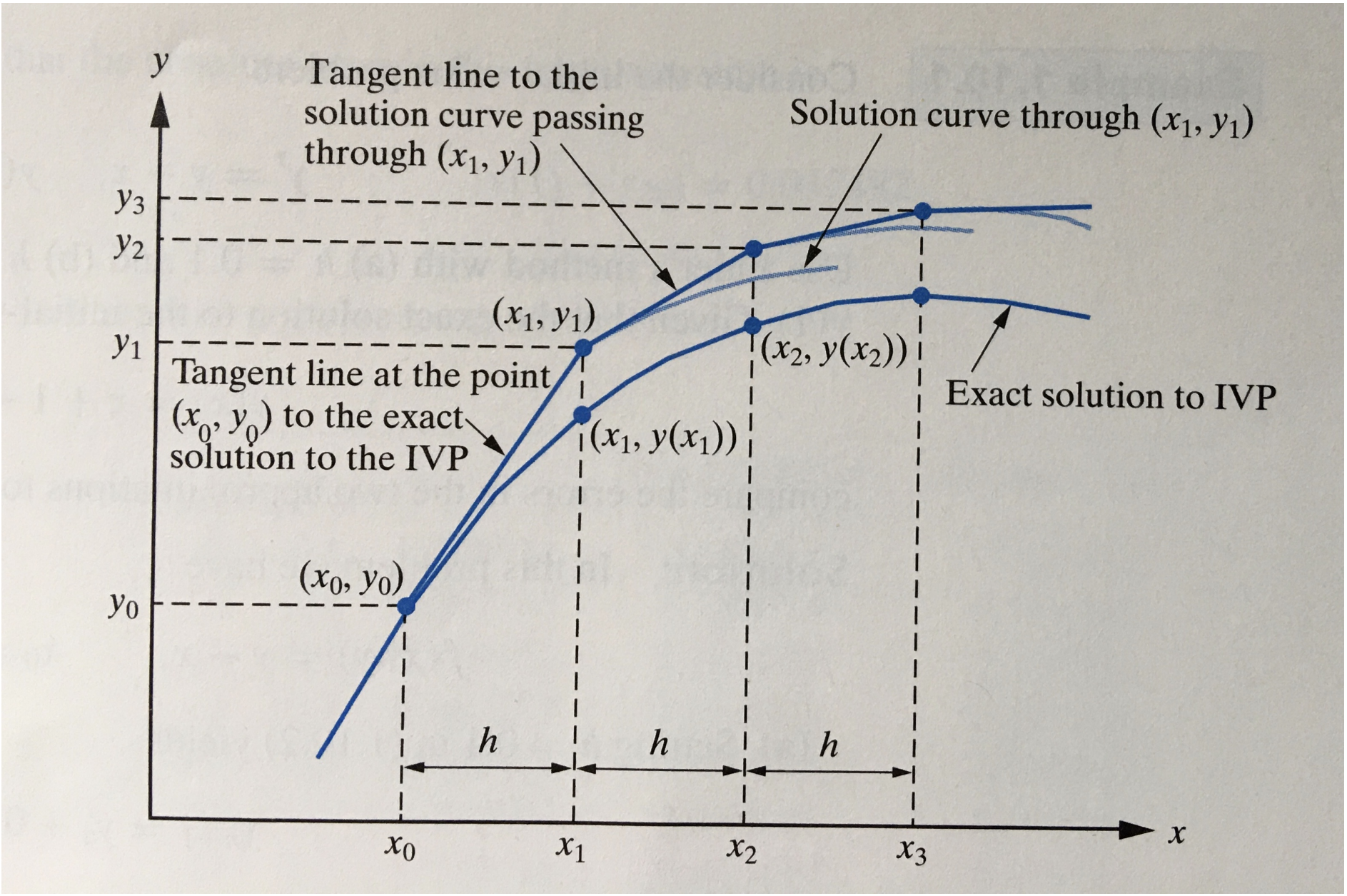
If we use a sufficient number of points, the by plotting the points (x_i, y_i) and joining them with straight-line segments, we are able to obtain an overall approximation to the solution curves corresponding to the solution of the given IVP.

We study two techniques, which are Euler's method and its modified, Heun's method (which is left as a self-study).

- **Euler's Method**.

First, suppose we wish to approximate the solution to the IVP (1) at $x = x_1 = x_0 + h$, where h is small.

The idea behind Euler's method is to use the tangent line to the solution curve through (x_0, y_0) to obtain such an approximation.


y
Tangent line to the solution curve passing through (x_1, y_1)
Solution curve through (x_1, y_1)
y_3
y_2
(x_1, y_1)
y_1
$(x_2, y(x_2))$
Exact solution to IVP
Tangent line at the point (x_0, y_0) to the exact solution to the IVP
$(x_1, y(x_1))$
(x_0, y_0)
y_0
h
h
h
x_0
x_1
x_2
x_3
x

The equation of the tangent line through (x_0, y_0) is

$$y(x) = y_0 + m(x - x_0),$$

where m is the slope of the curve at (x_0, y_0).

From (1) we have $m = f(x_0, y_0)$ and so

$$y(x) = y_0 + f(x_0, y_0)(x - x_0).$$

Setting $x = x_1$ in the last equation yields the so-called Euler approximation to the exact solution at x_1, namely

$$y_1 = y_0 + f(x_0, y_0)(x_1 - x_0),$$

which we write as

$$y_1 = y_0 + hf(x_0, y_0).$$

- Next, if we wish to obtain an approximation to the exact solution to the IVP (1) at $x_2 = x_1 + h$, then we can use the same idea, except we now use the tangent line to the solution curve through (x_1, y_1).

From (1) we have the slope of this tangent line is $f(x_1, y_1)$, so that the equation of the required tangent line is

$$y(x) = y_1 + f(x_1, y_1)(x - x_1).$$

Setting $x = x_2$ yields the approximation

$$y_2 = y_1 + hf(x_1, y_1),$$

where we have substituted for $x_2 - x_1 = h$ to the solution to the IVP at $x = x_2$.

- Continuing this process, we can determine the sequence of approximations

$$y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \ldots$$

to the equation to the IVP (1) at the points $x_{n+1} = x_n + h$.

In summary, Euler's method for approximating the solution to the IVP

$$y' = f(x, y), \quad y(x_0) = y_0$$

at the points $x_{n+1} = x_0 + (n+1)h \quad (n = 0, 1, \ldots)$ is

$$y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \ldots \tag{2}$$

Example

Consider the IVP

$$y' = y - x, \quad y(0) = \frac{1}{2}.$$

Use Euler's method with

(a) $h = 0.1$

(b) $h = 0.05$

to obtain an approximation to $y(1)$. Given that the exact solution to the IVP is

$$y(x) = x + 1 - \frac{1}{2}e^x,$$

compare the errors in the two approximations to $y(1)$.

Solution. In this problem we have

$$f(x, y) = y - x, \quad x_0 = 0, \quad y_0 = \frac{1}{2}.$$

(a) Setting $h = 0.1$ in (2) yields

$$y_{n+1} = y_n + 0.1(y_n - x_n).$$

Hence,

$$y_1 = y_0 + 0.1(y_0 - x_0) = 0.5 + 0.1(0.5 - 0) = 0.55$$
$$y_2 = y_1 + 0.1(y_1 - x_1) = 0.55 + 0.1(0.55 - 0.1) = 0.595.$$

Continuing this process, we can get the approximations listed in the table below (rounded to 6 decimal places).

n	x_n	y_n	Exact Solution	Absolute Error
1	0.1	0.55	0.547414	0.002585
2	0.2	0.595	0.589299	0.005701
3	0.3	0.6345	0.625070	0.009430
4	0.4	0.66795	0.654088	0.013862
5	0.5	0.694745	0.675639	0.019106
6	0.6	0.714219	0.688941	0.025278
7	0.7	0.725641	0.693124	0.032518
8	0.8	0.728205	0.687229	0.040976
9	0.9	0.721026	0.670198	0.050828
10	1.0	0.703129	0.640859	0.062270

We have also listed in the table above the values of the exact solution and the absolute value of the error.

In this case, the approximation to $y(1)$ is

$$y_{10} = 0.703129,$$

with an absolute error of

$$|y(1) - y_{10}| = |0.640859 - 0.703129| = 0.062270 \tag{3}$$

(b) When $h = 0.05$ Euler's method gives

$$y_{n+1} = y_n + 0.05(y_n - x_n), \quad n = 0, 1, \ldots, 19,$$

which generates the approximations given in the table below, where we listed only every other intermediate approximation.

n	x_n	y_n	Exact Solution	Absolute Error
2	0.1	0.54875	0.547414	0.001335
4	0.2	0.592247	0.589299	0.002948
6	0.3	0.629952	0.625070	0.004881
8	0.4	0.661272	0.654088	0.007185
10	0.5	0.685553	0.675639	0.009913
12	0.6	0.702072	0.688941	0.013131
14	0.7	0.710034	0.693124	0.016910
16	0.8	0.708563	0.687229	0.021333
18	0.9	0.696690	0.670198	0.026492
20	1.0	0.686525	0.640859	0.032492

So the approximations to $y(1)$ is

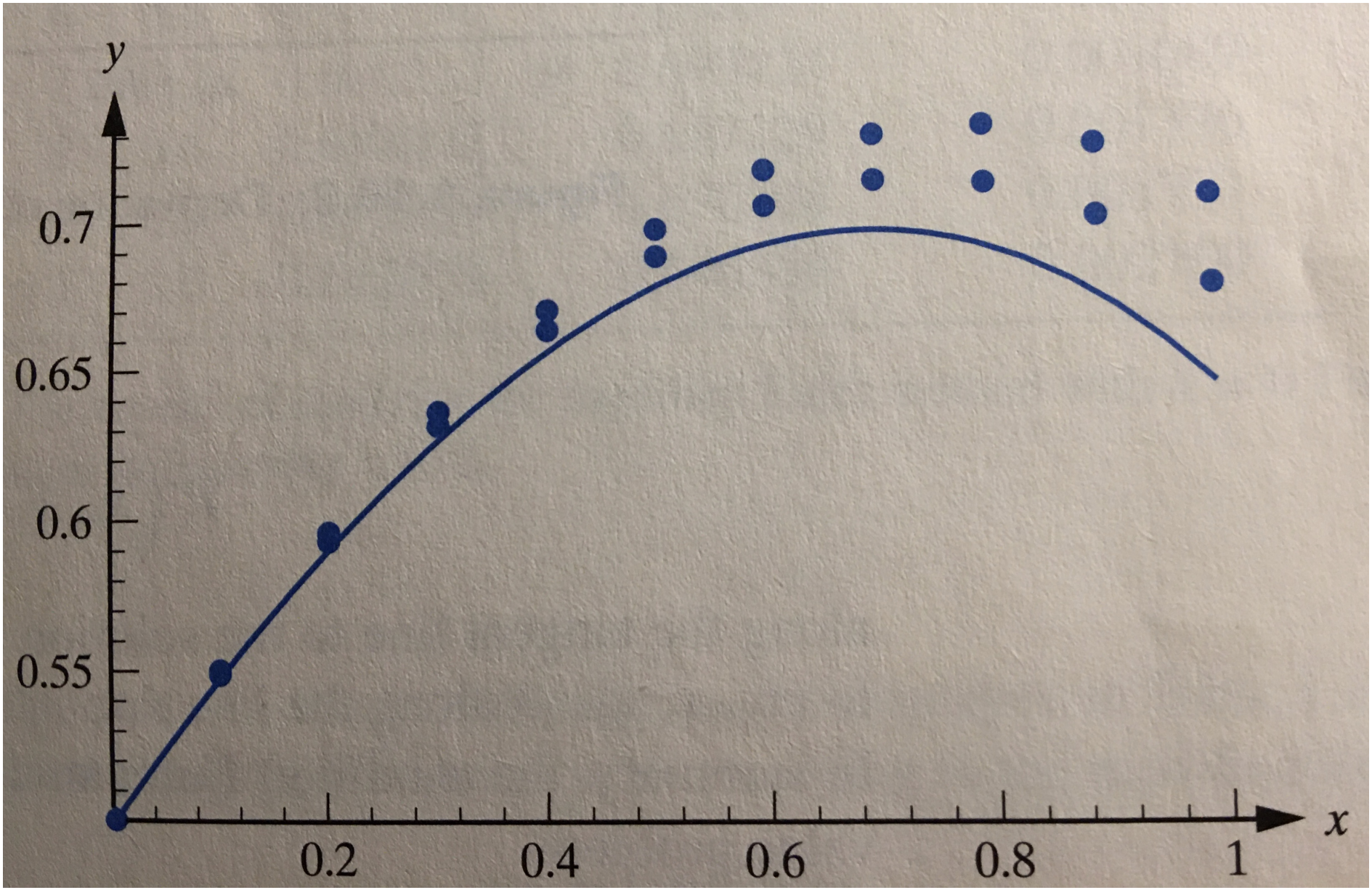
$$y_{20} = 0.686525,$$

and the absolute error in this approximation is

$$|y(1) - y_{20}| = |0.640859 - 0.686525| = 0.045666.$$

Comparing the two approximations, we see that the smaller steps size has led to a better approximation.

The exact solution to the IVP in the example above and the two approximations obtained using Euler's method is showed in the following figure.


y
0.7
0.65
0.6
0.55
0.2
0.4
0.6
0.8
1
x

Discussion. In the preceding example, we saw that halving the step size had the effect of essentially halving the error.

However, even then the accuracy was not as good as we probably would have liked.

Of course, we could just keep decreasing the step size (provided that we did not take h to be so small that round-off errors started to play a role) to increase the accuracy, but then the number of steps we would have to take would make the calculations very complicated.

$\Longrightarrow$ So a better approach is to derive method(s) that have a higher order of accuracy.

- **Modified Euler Method (Heun's method)** - **A self-study**.

END OF CHAPTER 3